ABSTRACT

HANDWRITTEN DIGIT RECOGNITION USING NEURAL NETWORKS (MNIST DATASET)

The Handwritten Digit Recognition project is a beginner-friendly Artificial Intelligence and Machine Learning project focused on recognizing handwritten digits from 0 to 9 using a Neural Network model.

The project uses the MNIST Handwritten Digits Dataset, which contains 70,000 grayscale images of handwritten digits. Each image has dimensions of 28 × 28 pixels. The dataset consists of 60,000 training images and 10,000 testing images.

The main objective is to develop an intelligent model capable of identifying and classifying handwritten numbers accurately. The project involves loading and exploring the dataset, preprocessing the images through normalization and reshaping where required, designing and configuring a Neural Network, and training the model while monitoring accuracy and loss.

The trained model is tested on unseen images, and its performance is evaluated using appropriate performance metrics. A simple digit recognition system is also deployed to allow users to input handwritten digits and view the predicted output.

This project provides practical understanding of Neural Networks, image classification, image processing, deep learning, model training, testing and performance evaluation.